PIOGLITAZONE

Heart failure
Bladder cancer & fractures

CHEMICAL NAME (RS)-5-{4-[2-(5-ethylpyridin-2-yl)ethoxy]benzyl}-1,3-thiazolidine-2,4-dione (hydrochloride)
CLASS Thiazolidinedione (glitazone) — PPAR-γ agonist
TRADE NAMES Actos · Glustin · Pioz · Competact (with metformin)

THERAPEUTIC USES / INDICATIONS

- Type 2 diabetes — add-on or alternative when metformin unsuitable, especially with marked insulin resistance.
- Non-alcoholic steatohepatitis (NASH) with diabetes (off-label).

ADMINISTRATION

- **PO** once daily with or without food.

DOSAGES

- **Adult:** 15–30 mg once daily; max 45 mg.
- With insulin: max 30 mg (fluid retention).
- Review after 3–6 months; stop if inadequate response.

PHARMACOKINETICS

- **Onset:** glucose falls after 2 weeks; full effect 8–12 weeks.
- Metabolised in liver (CYP2C8) to active metabolites.
- Excreted in bile and urine. **t½:** 3–7 h (metabolites 16–24 h).

PHARMACODYNAMICS

- Activates **PPAR-γ** in adipose, muscle and liver → ↑ insulin sensitivity, ↑ glucose uptake, ↓ hepatic glucose output.
- Does not cause hypoglycaemia alone; ↑ HDL.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Weight gain, oedema (fluid retention).
- Headache, upper respiratory infection.
- Visual disturbance (macular oedema).

ADVERSE EFFECTS (SERIOUS)

- Heart failure (fluid retention).
- Bladder cancer (long-term).
- Bone fractures (especially women, distal limbs).
- Hepatotoxicity (rare).

CONTRAINDICATIONS & CAUTIONS

- Heart failure (any grade) or history.
- Hepatic impairment; DKA.
- Active or history of bladder cancer; uninvestigated haematuria.
- Pregnancy, breastfeeding.

KEY INTERACTIONS

- **Insulin:** ↑ oedema and heart failure.
- **Gemfibrozil (CYP2C8):** ↑ levels.
- **Rifampicin:** ↓ levels.

NURSING MANAGEMENT

- Baseline LFTs and periodically.
- Monitor weight, oedema, breathlessness (heart failure).
- Ask about haematuria.
- Assess fall/fracture risk.

PATIENT EDUCATION

- Full effect takes 2–3 months.
- Report rapid weight gain, ankle swelling, breathlessness, blood in urine, dark urine.
- Women: may restore ovulation — use contraception.

KEY POINTS

- PPAR-γ agonist; insulin sensitiser.
- Contraindicated in heart failure.
- Slow onset (weeks).
- Bladder cancer, fractures, oedema.